



CHAP

The Collaborative Human-Agent Protocol

One shared, verifiable record of how people and AI agents work together.

● open protocol

arXiv 2606.09751

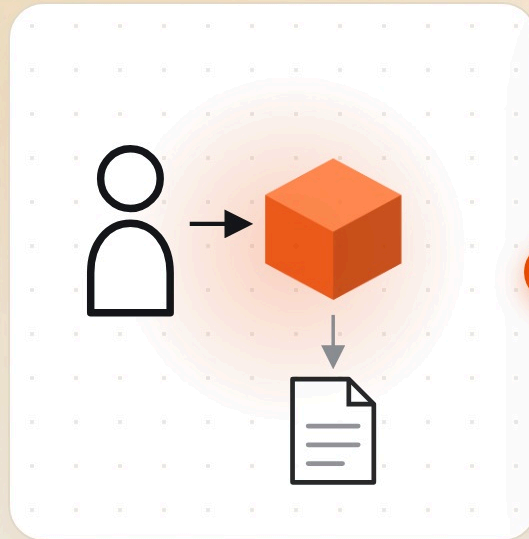
github.com/BrightbeamAI/chap

live on npm + PyPI

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From one model to **teams of humans and agents**

GENERATIVE AI

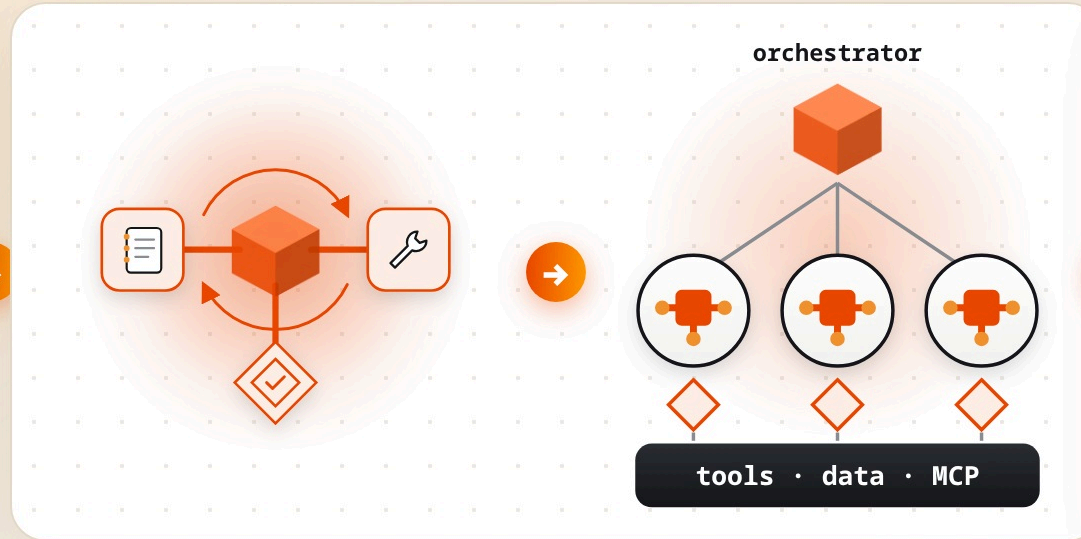


a question, an answer

Wave I

one human, one model, one answer

AGENTIC AI

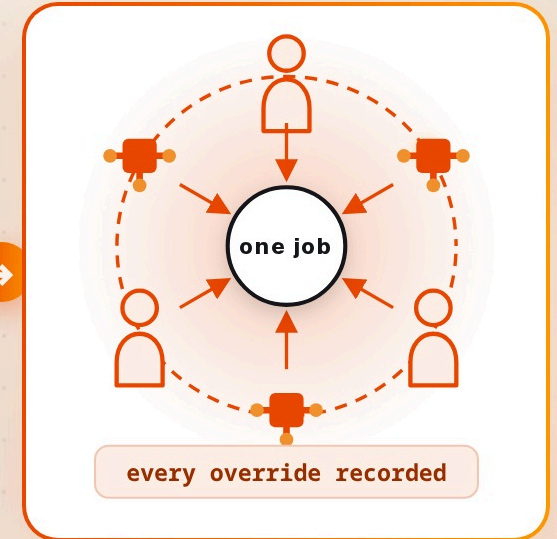


one agent runs the plan

Wave II

an orchestrator, many models, many tools

HUMAN-AGENT SYMPHONY



many humans, many agents,
shared work

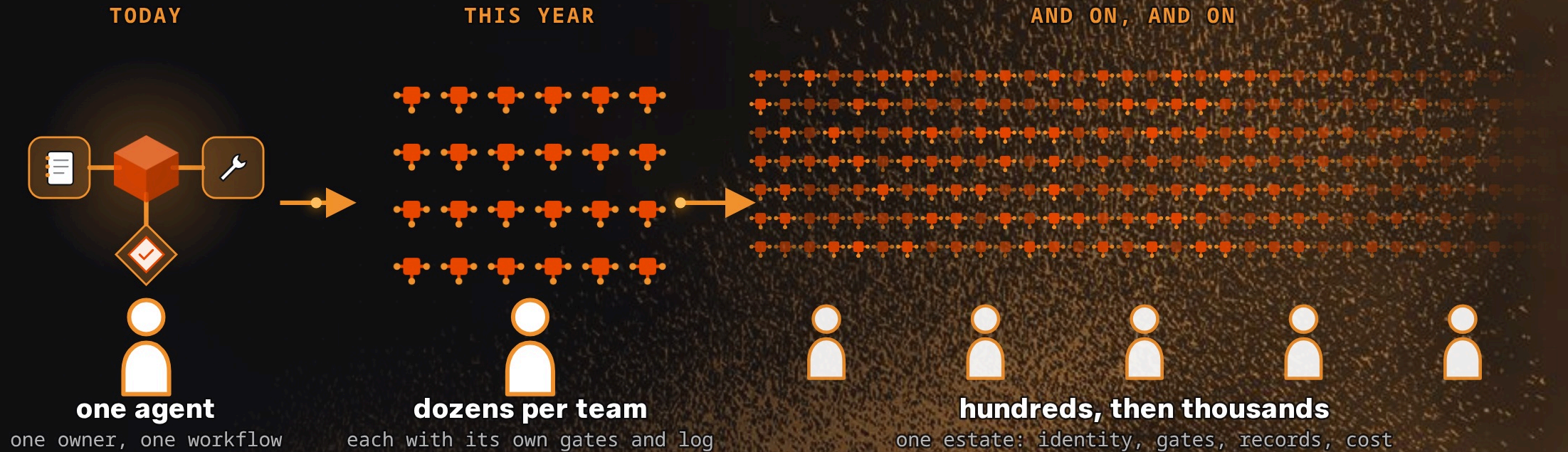
Wave III · the symphony

many humans + many agents,
one problem

In regulated work this becomes a regulatory problem that gets re-engineered on every single project.

— THE SCALE THAT IS COMING

One agent today. **Thousands to run.**



WHAT HAS TO SCALE

the governance

identity, gates, records and cost per agent

— THE ENGINEERING SHIFT

One piece of work, **many hands on it**



who did what?

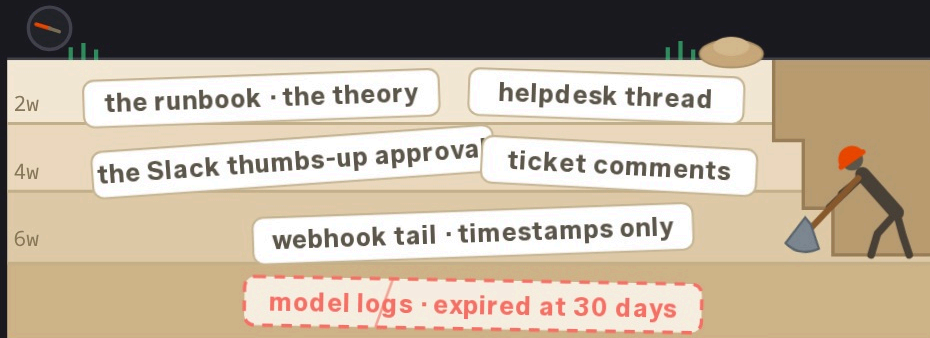
in **what order** ?

on **whose authority** ?

— THE OPERATIONAL COST

30 seconds to query, or 45 minutes to reconstruct

TODAY · THE DIG



~45 minutes

of careful digging · 4 tools · half the shards missing

WITH ONE SHARED RECORD · THE DISPLAY CASE

the ask
task.create

the draft
artefact

the decision
decide.override

chained by fingerprints · tamper-evident

~30 seconds

one query · verifiable years later

45 minutes of archaeology for a ten-second decision. CHAP turns the dig into a query.

Human judgement remains **the highest-value signal**



Evidence

proof that human oversight actually happened



Training data

the exact correction that would fix the agent



Accountability

who decided, under which policy, and why



Today it vanishes into

chat threads

ticket comments

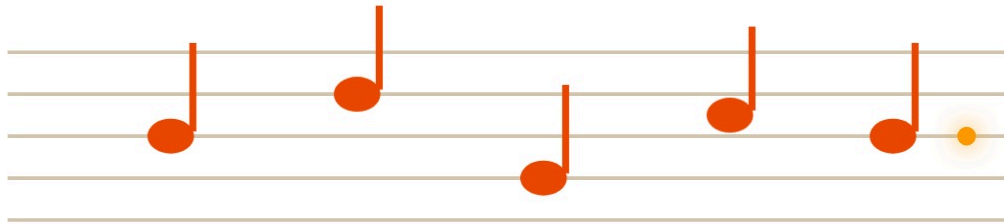
application code

tribal memory

Approval. Correction. Override. Abstention. Escalation. Every one of them is worth capturing.

Only what is written **can be replayed, verified or improved**

The oldest lesson in collaboration



For a thousand years, music **died with its players**. Then notation.

Computing keeps re-learning it

- TCP/IP** machines got a shared language
- HTTP** documents got one
- OIDC** identity got one
- MCP** agent-to-tool got one
- A2A** agent-to-agent got one
- CHAP** **human-agent work gets its written form**

CHAP is **musical notation for the human-agent symphony**.

Shared work needs **a shared language**

WHAT A RECORD HAS TO SURVIVE



teams, tools and agents change · the record stays

 **explicit**

written down as it happens

 **observable**

visible to whoever is accountable

 **portable**

travels across tools and teams

 **shared**

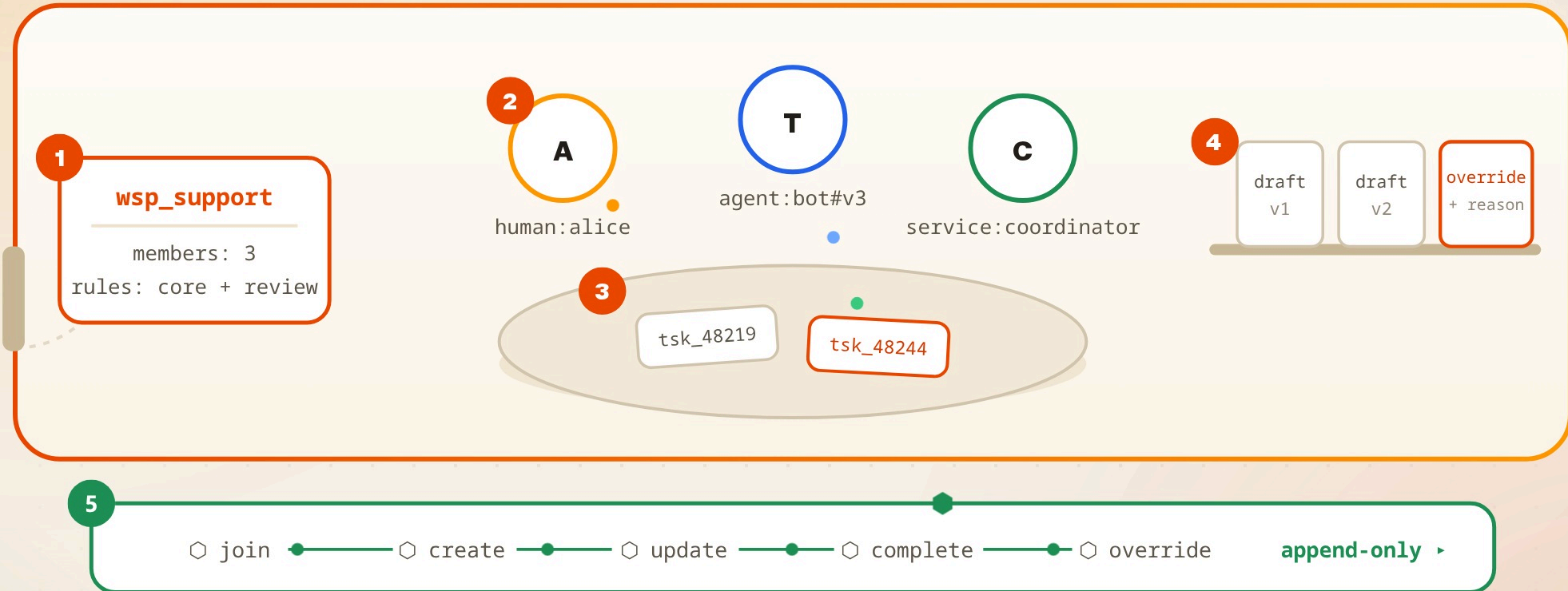
one vocabulary for everyone

CHAP ANSWERS ONE QUESTION

What are the smallest building blocks needed to describe **how people and agents work together?**

— THE CONCEPTUAL MODEL

Inside a workspace: **five nouns**



1 Workspace · the room, rules on the door **2 Participants** · typed badges **3 Tasks** · the tickets **4 Artefacts** · what's made

5 Evidence log · the memory

— THE MANDATORY CORE

Seven verbs, **all mandatory**



`workspace.describe`

**Read the placard on
the door**



`participant.join`

Sign the visitor book



`participant.leave`

Sign out



`task.create`

**Hand over the job
ticket**



`task.update`

The progress call



`task.complete`

The delivery note



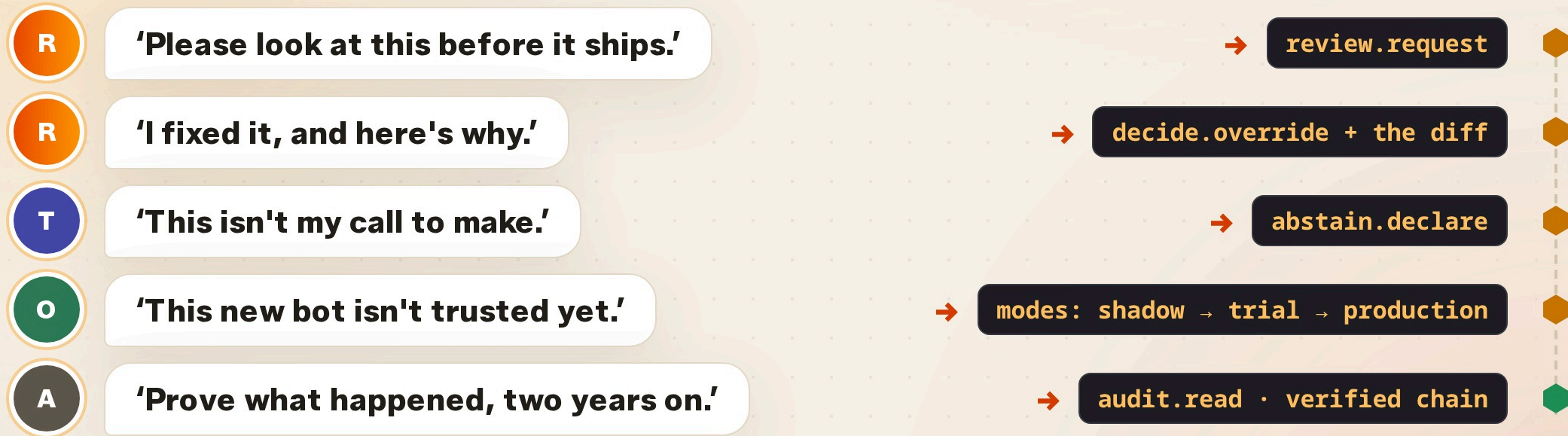
`audit.read`

**Replay the flight
recorder**

= Core

everything mandatory

Five sentences **every team needs on the record**

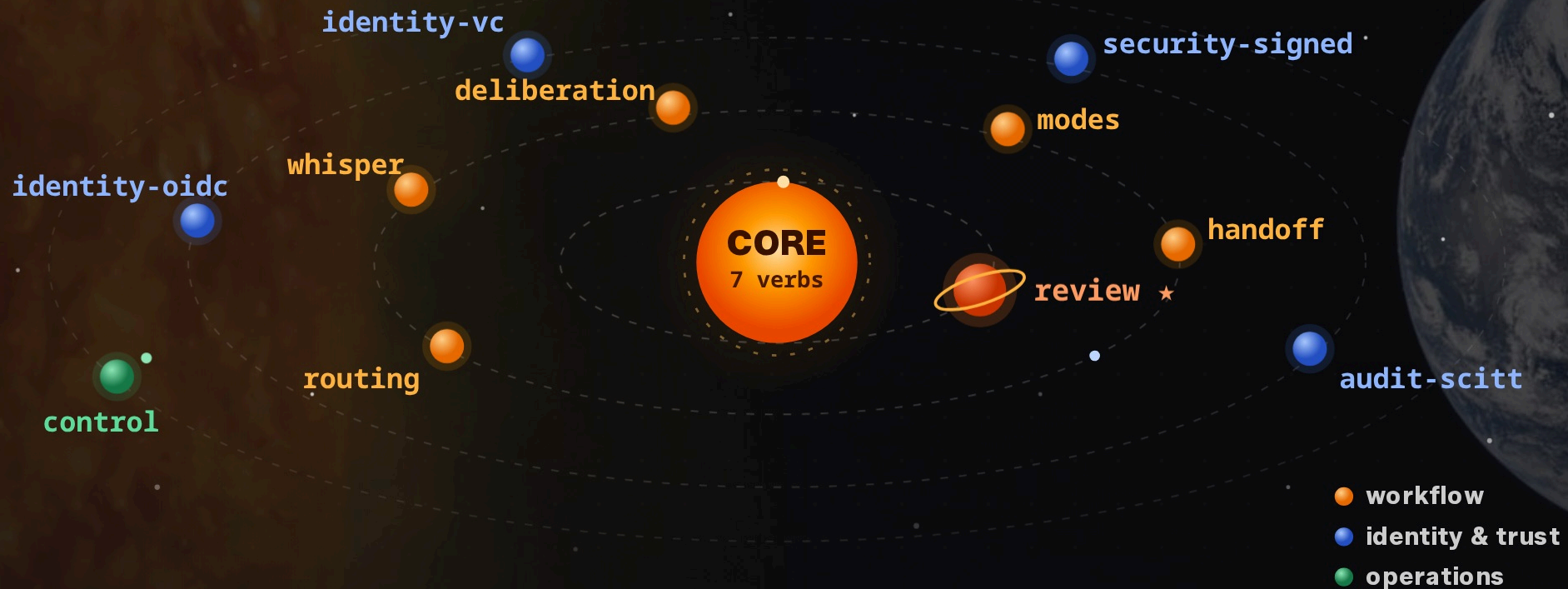


R the reviewer T the bot itself O ops lead A the auditor

Seven verbs are the Core. **The optional profiles let you say the rest.**

— ELEVEN OPTIONAL PROFILES

A small core. Everything else orbits it.



Versioned, optional, advertised on the door. **Add a planet when you feel its pull.**

Everything anyone says **is one envelope**



```
{
  "jsonrpc": "2.0",    same plumbing as MCP
  "id": "01HZ9YWQ...",
  "method": "task.create",
  "params": {
    "workspace": "wsp_support",    which room
    "from": "human:alice@acme.io", who speaks
    "to": "agent:triage-bot#v3",   to whom
    "ts": "2026-05-17T09:14:22Z", when
    "kind": "draft_response",
    "input": { "ticket_id": "INC-48219" }
  }
}
```

typed names **human: · agent: · service:** – and agents carry versions

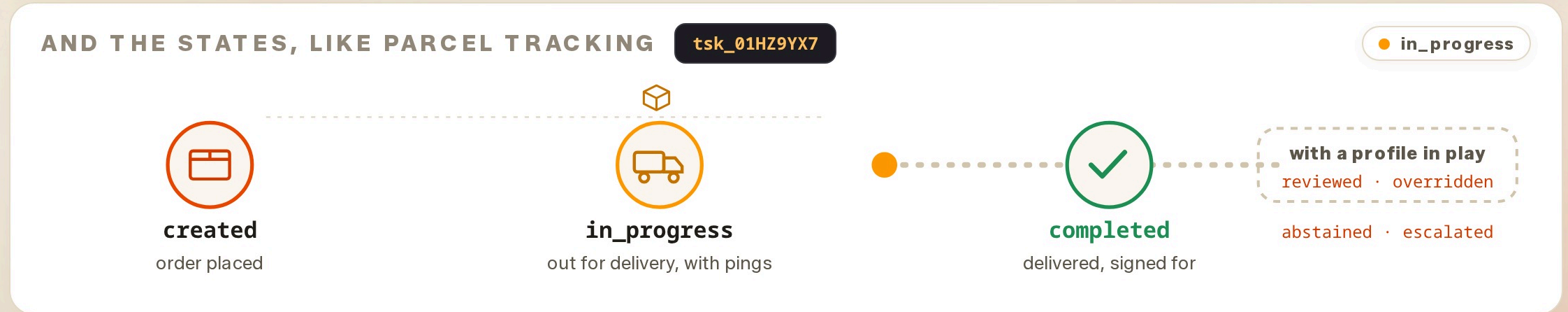
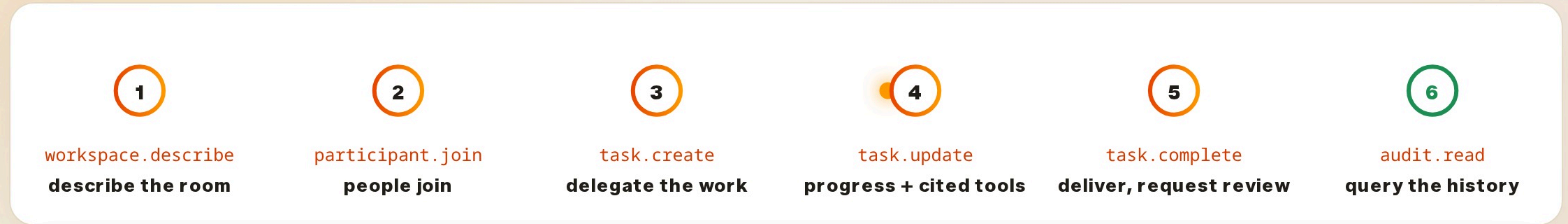
4 required fields **room · speaker · addressee · time.**
Everything else is optional.

boring on purpose If you've written an MCP client, **you already know this wire.**

Boring is what interoperability feels like. The new ideas live in the fields rather than the plumbing.

— THE SHAPE OF A SESSION

A task, tracked like a parcel



— WHERE THE VALUE COMPOUNDS

Human judgement compounds **into digital intelligence**



Every human decision becomes **structured evidence**.

The evidence shows **where agents perform, fail, and still need people**.

Captured at source, as it happens.

A support ticket, **from draft to decision**

chap · wsp_support_triage

rules: core + review



● LIVE

SCENARIO a customer asks "where is my order?" · **alice** supervises · **triage-bot** drafts · six envelopes, one chain

1 A human delegates the task

```
{ "method": "task.create",  
  "params": { "from": "human:alice@acme.io",  
              "assignee": "agent:triage-bot#v3.2",  
              "kind": "draft_response",  
              "input": { "ticket_id": "INC-48219" } } }
```

1 Delegate



2 Cited tool



3 Draft



4 Review



5 Human decides



6 Recorded

A human delegates. The ticket carries structured input and a named assignee, and the room checks that assignee is actually a member before it accepts the task.

Step ›

◀ Back

Reset

step 1 / 6

rules: core + review



● **LIVE**

2 The agent uses a cited tool

1 Delegate

2 Cited tool

3 Draft

4 Review

5 Human decides

6 Recorded

Step ›

Reset

step 2 / 6

A support ticket, from draft to decision

● LIVE

3 The agent produces a draft

"We ~~sincerely apologise for the extreme delay~~... your order is in transit and will arrive within three business days. We ~~deeply regret any inconvenience caused~~..."

6 Recorded

step 3 / 6

4 The agent requests review

1 Delegate → 2 Cited tool → 3 Draft → 4 Review → 5 Human decides → 6 Recorded

Step ›

Reset

step 4 / 6

A support ticket, from draft to decision

rules: core + review

SCENARIO a customer asks **"where is my order?"** · **alice** supervises · **triage-bot** drafts · six envelopes, one chain

THE HUMAN'S VERSION

reason: "Too apologetic for a routine tracking question."

6 Recorded

step 5 / 6

● LIVE

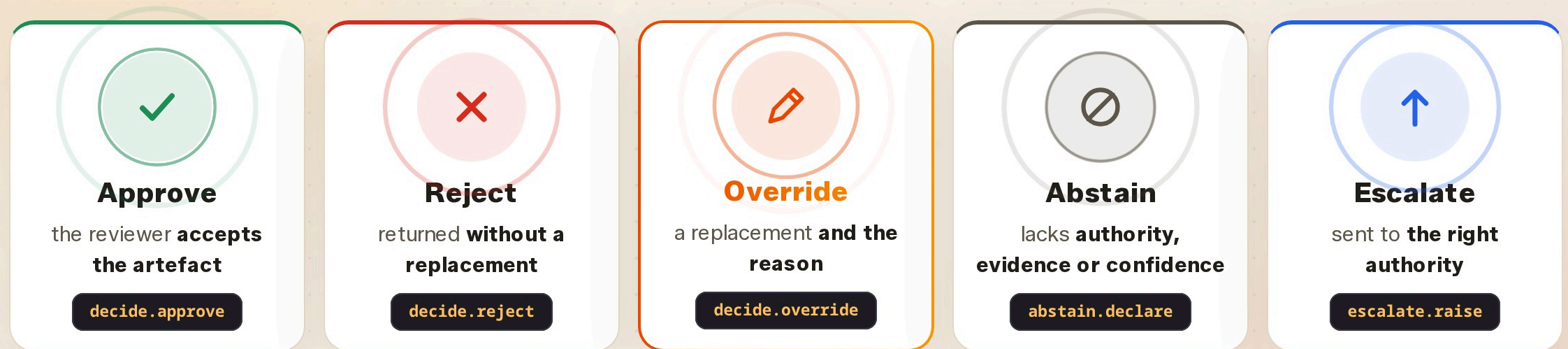
6 CHAP records the decision and its reasoning

1 Delegate → 2 Cited tool → 3 Draft → 4 Review → 5 Human decides → 6 Recorded

step 6 / 6

— THE FIVE REVIEW ACTIONS

Each decision says **something different**



Five buttons for the human. **Five different signals.**

— INSIDE AN OVERRIDE

An override records **the correction and the reasoning**

```
{
  "method": "decide.override",
  "params": {
    "task_id": "tsk_01HZ9YX7",
    "from": "human:alice@acme.io",
    "intent_preserved": true,
    "diff": [{ "op": "replace", "path": "/body",
              "value": "Thanks for following up..." }],
    "rationale": "Too apologetic for a
                  routine tracking question.",
    "tags": ["tone-softened"],
    "policy_refs": ["support-tone-v2"]
  },
  "evidence": { "prev_hash": "sha256:9f3c..." }
}
```

±/ **diff**

a **standard JSON Patch** – a machine can re-apply the fix

99 **rationale**

the reason, **in the human's own words**

🏷 **tags**

the team's own vocabulary – **the axis you aggregate on**

🎯 **intent_preserved**

true = same decision, better words · **false** = a different decision

∞ **prev_hash**

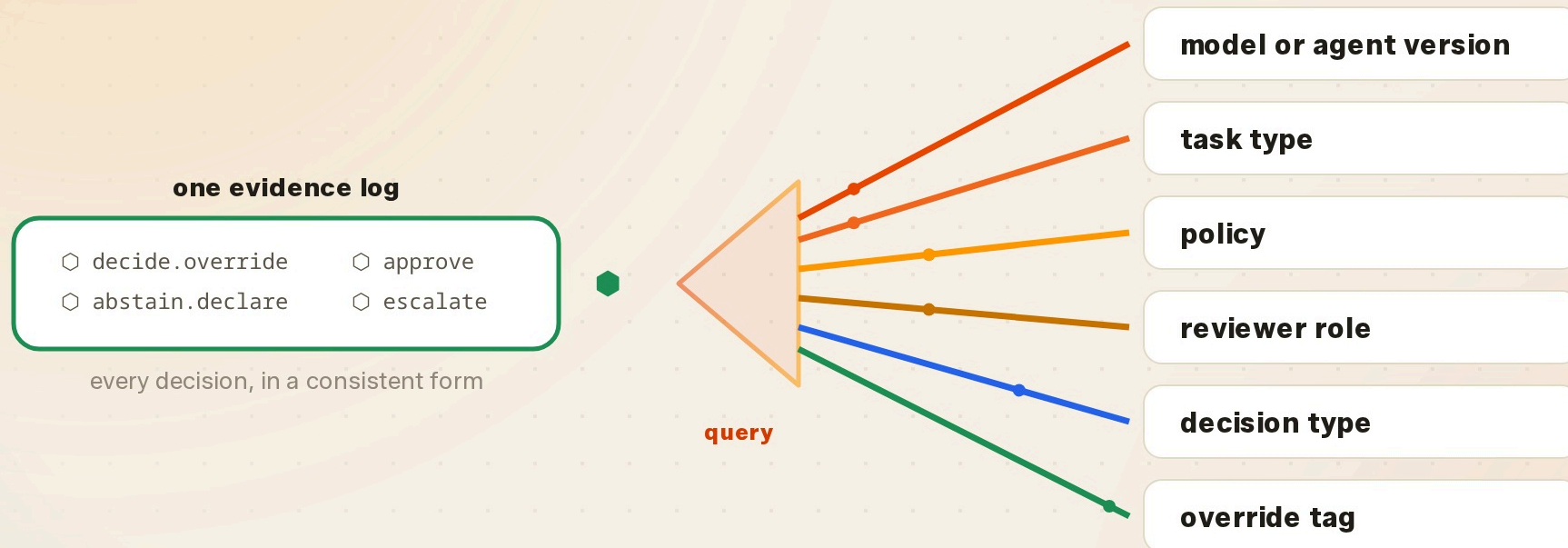
chained to the draft it corrects: **a record rather than a memory**



A corrected document shows the result. This shows what changed, who changed it, and why.

— FROM ONE DECISION TO THE WHOLE RECORD

The evidence log becomes **an intelligence dataset**



Cut it any way you like. **The log already holds the answer.**

Different patterns point to **different fixes**



Refining overrides **climb**



retrieval

templates

phrasing

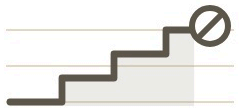


Substituting overrides **climb**



missing context

ambiguous policy



Abstentions **climb**



authority boundary

review routing

The backlog stops being a guess. **The evidence names the lever.**

— FROM EVIDENCE TO AUTONOMY

Evidence is what **allows autonomy**



1

Who can act

who may do this work at all



2

Who must review

and how deeply, for work of this risk



3

Who is responsible

a named person rather than a team inbox



4

What evidence earns autonomy

what results earn independence



5

How it is withdrawn

paused or removed at any moment

CHAP makes this **map of authority visible through the work itself.**

— TWO FORMS OF HUMAN INVOLVEMENT

Small questions get a timer. Big decisions get a vote.

whisper

'the tap on the shoulder'



'Form says the 17th, photos say the 19th. Which date?'



✓ answered in 12 s

🕒 If it lapses, the safe default fires

Either way, the agent moves on.

whisper.ask → whisper.answer · safe default

deliberation

'the jury room'



rule: `weighted_vote ≥ 2.0`

other rules: `all_approve` · `quorum:N` · `with veto`

✓ handler 1.0

✓ senior 1.5

✗ veto? stays in the log, with its reasoning



0

bar: 2.0

votes: 2.5 · passed

The outcome is computed rather than remembered. The minority view stays in the log.

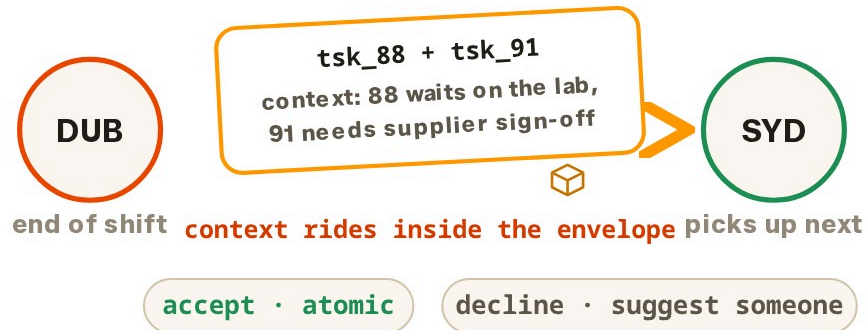
deliberate.open → comment → vote → close

— WORK THAT MOVES BETWEEN PEOPLE AND AGENTS

Hand-off without losing context

handoff

'the shift handover'

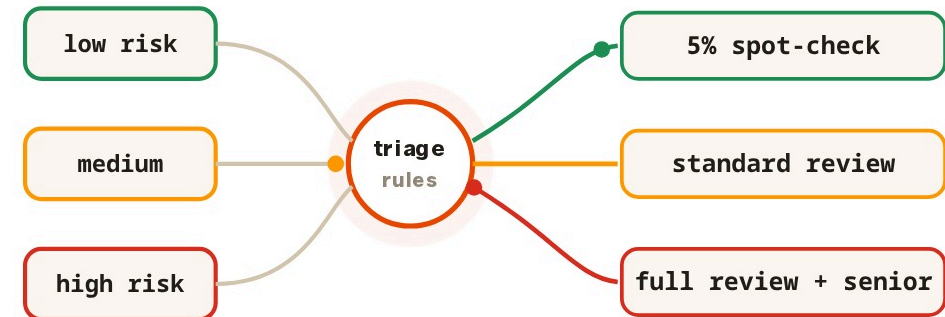


The chart travels with the patient.

handoff.propose → handoff.accept / handoff.decline

routing

'the triage desk'



Risk decides who reviews, and how deeply. The policy gets audited as well as the people.

task.route · review.depth · escalate.auto

— SHADOW, TRIAL, PRODUCTION

A new agent **earns trust in stages**



shadow works beside the old flow · its drafts stay internal

trial output ships · a human reviews **every single piece**

production **trusted** · reviews drop to risk-based sampling ·
demotion is one call away

the brake pause · snapshot · roll back · clamp · one audited call
each

Promotion is argued from the log.

A claim that can answer **the assurance questions**

chap · wsp_claims_review

rules: core + review + whisper + routing + deliberation + signed + scitt

● LIVE

**FOUR
QUESTIONS**

Who sent this? · Is that really the reviewer? · Is the signer qualified? ·
Was the log rewritten? ·

**identity, credentials, signatures and the evidence
chain answer all four**

000-001

participant.join
n ×2

human:claims-
handler ·
agent:claims-
bot#v4.1

○ chain start

002

task.create

CLM-20441 · hint:
criticality high

○ sealed

003

task.route +
review.depth

→ claims-bot · review:
full

○ decision logged

004-005

whisper.ask /
answer

'which date of loss?'
→ photo date · 12 s

○ answered in time

006

task.complete

€4,120 draft · cites
its MCP tool call

○ tool call cited

The delivery note: a €4,120 settlement draft with 0.83 confidence. The bot's **MCP tool call to the policy database is cited inside the record**: the server, the call, and fingerprints of its inputs and outputs.

Step ›

◀ Back

Reset

beat 5 / 10

A claim that can answer **the assurance questions**

chap · wsp_claims_review

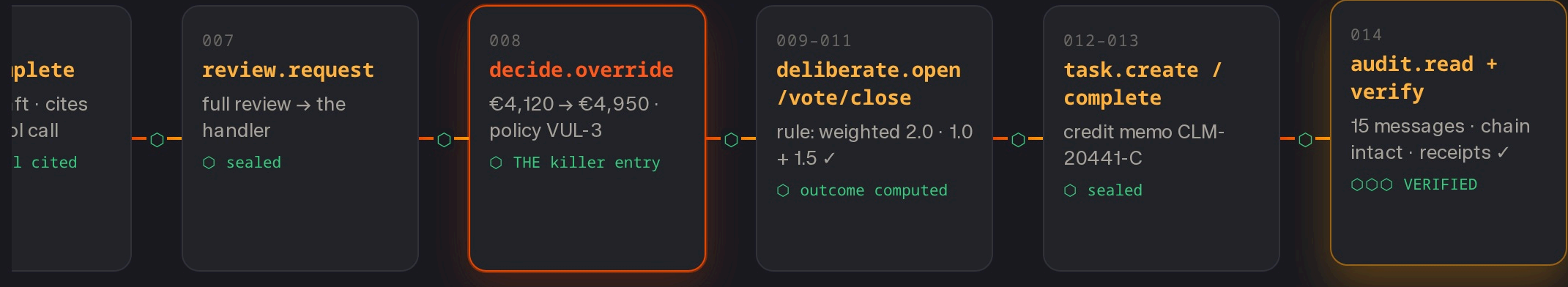
rules: core + review + whisper + routing + deliberation + signed + scitt

● LIVE

**FOUR
QUESTIONS**

Who sent this? · Is that really the reviewer? · Is the signer qualified? ·
Was the log rewritten? ·

**identity, credentials, signatures and the evidence
chain answer all four**



Six months later, the regulator asks. One replay returns all fifteen messages; the fingerprint chain re-verifies end to end; public-log receipts prove nothing was rewritten. **The 45-minute scavenger hunt is now a 30-second answer.**

Step ›

◀ Back

Reset

beat 10 / 10

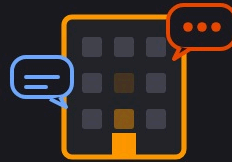
The same seven verbs, **from a laptop to a factory floor**



One developer

a code-review bot that **learns from its overrides**

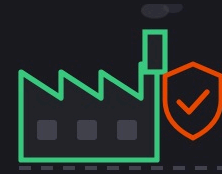
core + review



A support team

forty drafts a day, **risk-routed**

+ routing + whisper



A regulated line

a 3am release, **signed and replayed**

+ credentials + signed + public log

EU AI Act · Art 14 & 12

NIST AI RMF

GMP Annex 11

DORA

SOX-404

HIPAA

CHAP produces the evidence these regimes ask for. The compliance case stays yours to make.

— BEYOND ONE COMPANY

A protocol only becomes infrastructure **when others build on it**



Open specification

every method and profile, in public

CC-BY 4.0



Working reference code

runnable implementations you can
start today

Apache 2.0



Compatibility checks

21 ready-made checks: does your
build speak CHAP?

ready to run

● v0.2 public draft



1.0

Independent implementations will close the gap.

That requires engineers beyond Brightbeam to test and challenge CHAP.



Five minutes to **a running server**

1 Clone
the public repository

2 Install
a single package

3 Run
the reference server, locally

4 Send the first envelope
plain curl is all you need

```
$ git clone https://github.com/BrightbeamAI/chap
$ cd chap/reference/core && npm install
$ npm run start:demo
CHAP Core → http://localhost:8080/chap
```

```
# the first envelope: a reviewer joins the room
curl -s -X POST http://localhost:8080/chap \
  -H 'Content-Type: application/json' \
  -d '{"jsonrpc": "2.0", "id": "1",
    "method": "participant.join",
    "params": {"workspace": "wsp_demo",
               "from": "human:alice@acme.io",
               "role": "reviewer"}}'
```

Join the agent, create a task, read the audit. The whole loop.

— PRACTICAL COMPOSITION

It works with **what is there already**

Running MCP?

ships **as an MCP server**: every verb a tool

```
tools: chap.task_create · chap.decide_override ·  
chap.audit_read ...
```

Running A2A?

ships **as an A2A agent**: every verb a skill

```
AgentCard.skills: ["task.create", "review.request", ...]
```

Using an agent framework?

Bridges wrap the **human-input boundary** in about three lines.

[LangGraph](#)[LlamaIndex](#)[Pydantic AI](#)[AG2](#)[Google ADK](#)

BUILT ON EXISTING STANDARDS JSON-RPC 2.0 · JSON Patch · OIDC · W3C VC · IETF SCITT



— THREE ASKS

01

Try it

02

Share it

03

Build on it

'The Human-Agent Symphony will not emerge simply because models improve. It emerges when the collaboration layer becomes **explicit, observable, portable and shared**. CHAP is a proposal for that layer.'



Thank you.

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github.com/BrightbeamAI/chap

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